

Colorado Risk-Limiting Audits: Three Years In

Colorado Department of State
Voting Systems Team

Introduction

Since the 2017 Coordinated Election, Colorado counties have conducted Risk-Limiting Audits (RLA) to verify the outcomes of elections to a high level of statistical confidence. Before counties could perform this cutting-edge election audit, voting systems capable of exporting cast vote records (CVRs) had to be developed by voting system providers, tested and certified by the Secretary of State, and purchased by counties. Having voting systems capable of exporting CVRs was just the start, Colorado then also had to develop a first of its kind software application to facilitate the audit, and write election rules which took a concept and made it a reality. Previous to any of the above happening, limited number of counties performed pilots and worked with the Secretary of State's office to create a foundation for the future. Now that five audits have been performed in various types of elections, we believe it is a good time to take a look back see how things have changed since 2017.

Types of Audits

- Ballot Polling Audit

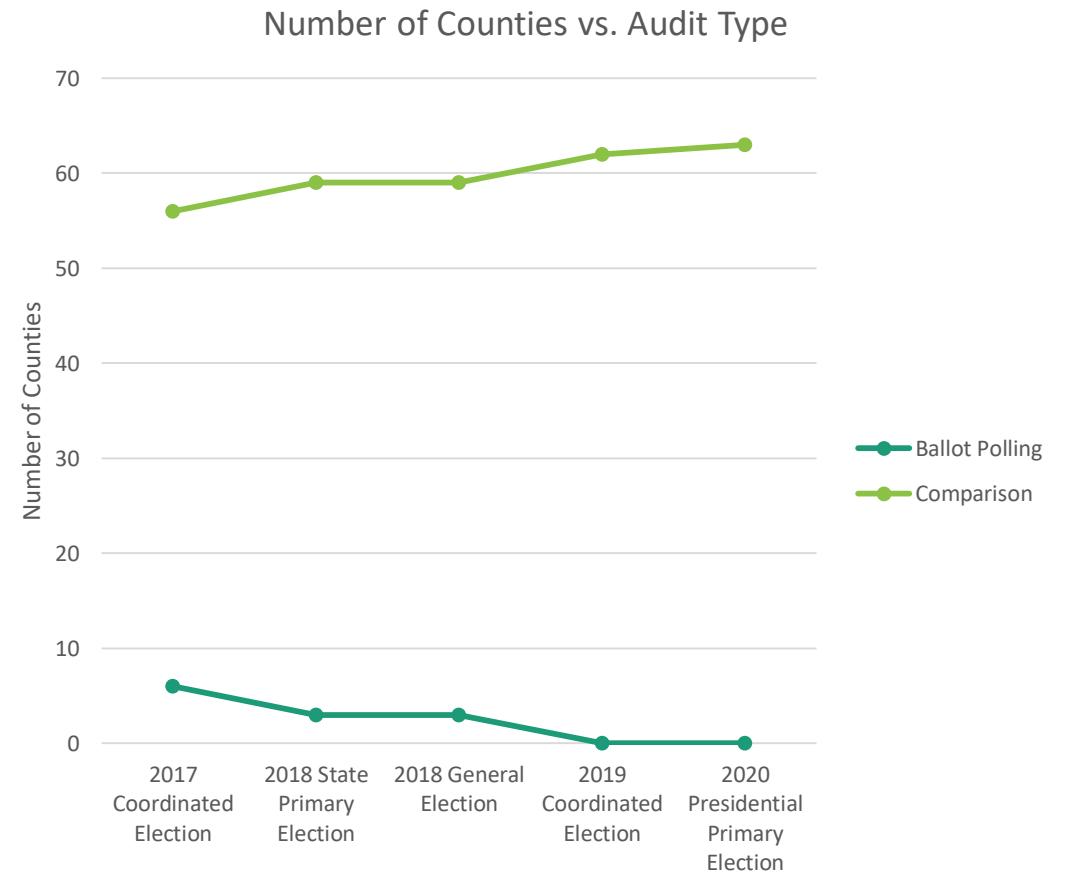
- Similar to an exit poll, the county is instructed to locate and retrieve a number of specific ballots based on the margin between the winner and loser of a race. The county then reports the vote totals for the targeted race to the state. If the results of the sample confirm the outcome, the audit concludes. If the sample does not confirm the outcome, an additional round of auditing is required.

- Ballot-Level Comparison Audit

- The results of each ballot are compared against how the voting system tabulated that individual ballot as shown in the cast vote record file. The comparison is done at the ballot level, rather than the sample level, resulting in fewer ballots being required to confirm the outcome. This method requires more sophisticated software.

Type of Audits Over time

Due to a number of circumstances (e.g., ability to procure a new voting system, expertise required to perform an audit) some counties performed ballot polling audits, and others performed comparison audits. Gradually, Colorado has shifted to all counties performing comparison audits, except San Juan county, which hand tabulates their ballots.



Baseline Statistics (Comparison Audits)

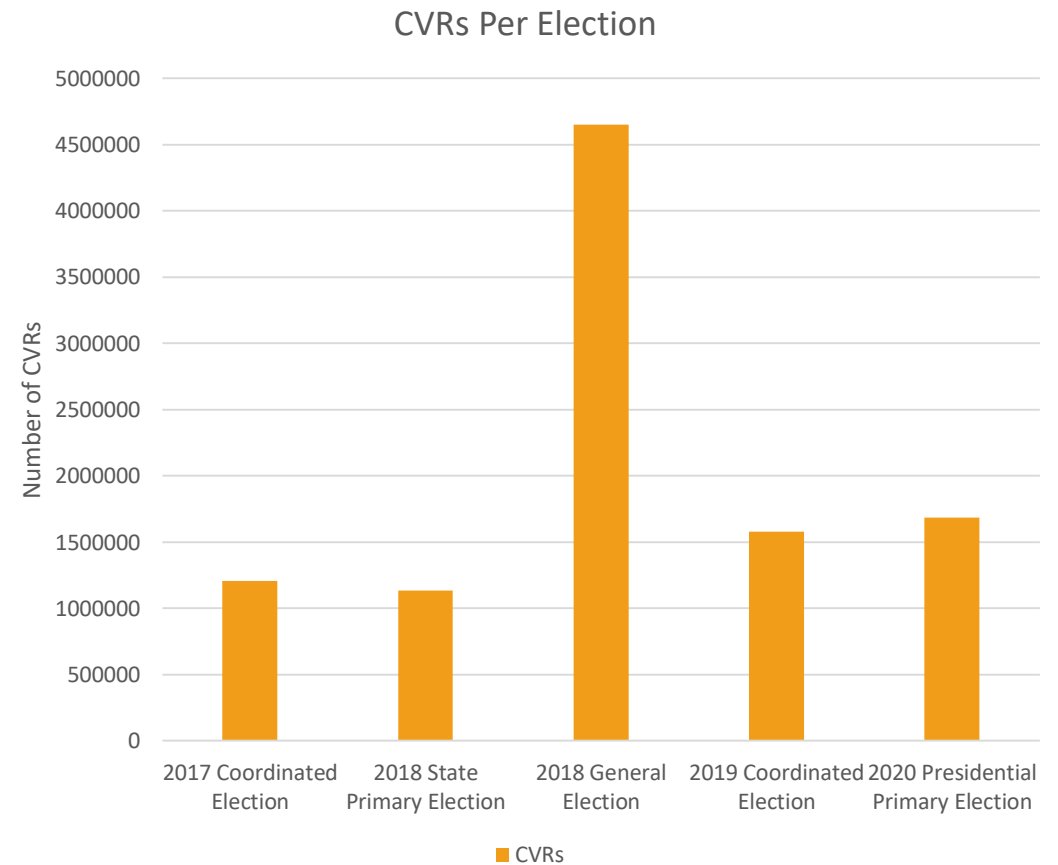
Here we present some statistics that show the breadth of the comparison audit over time, and how counties have become more proficient at performing the audit using the following metrics

- Number of CVRs uploaded
- Number of ballot cards audited
- Total number of discrepancies
- Discrepancies per card

These statistics are taken from all of the comparison audits since Colorado began performing RLAs after the 2017 Coordinated Election.

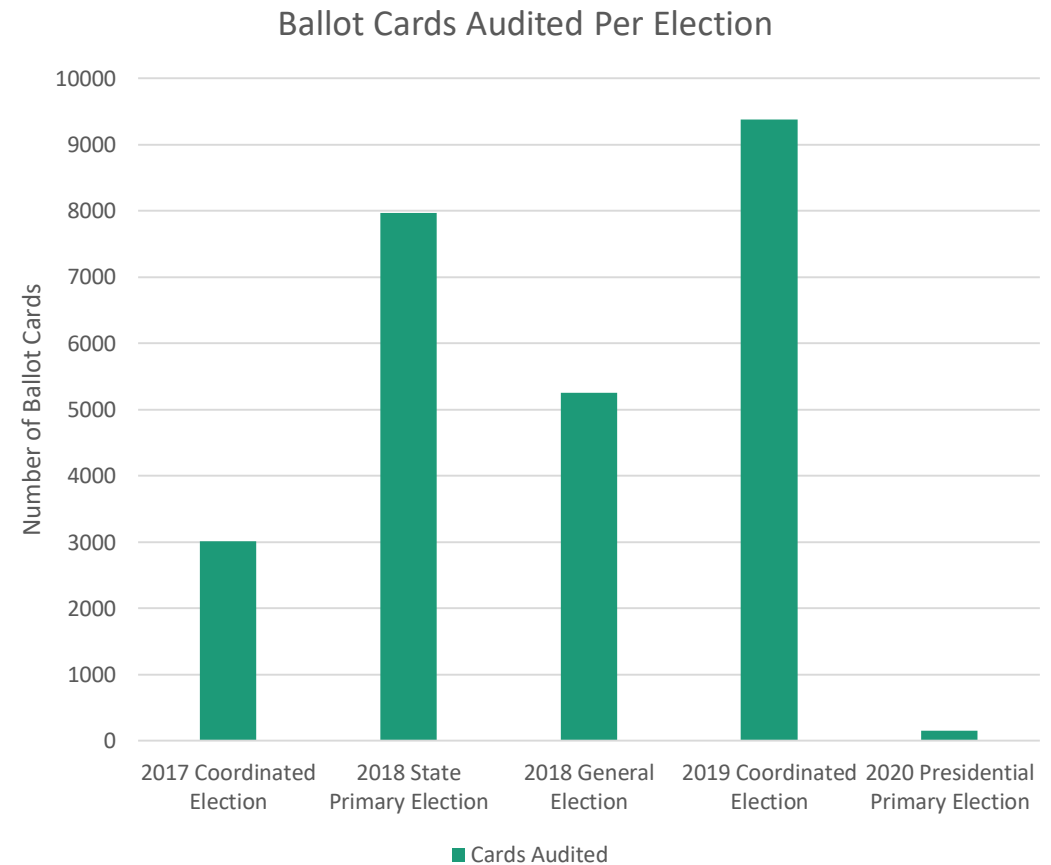
Number of CVRs Uploaded

A CVR (cast vote record) is a record of how the voting system tabulated a ballot card. The total number of CVRs equals the total number of ballot cards scanned by the voting system(s). This number usually varies by type of election because of voter interest.



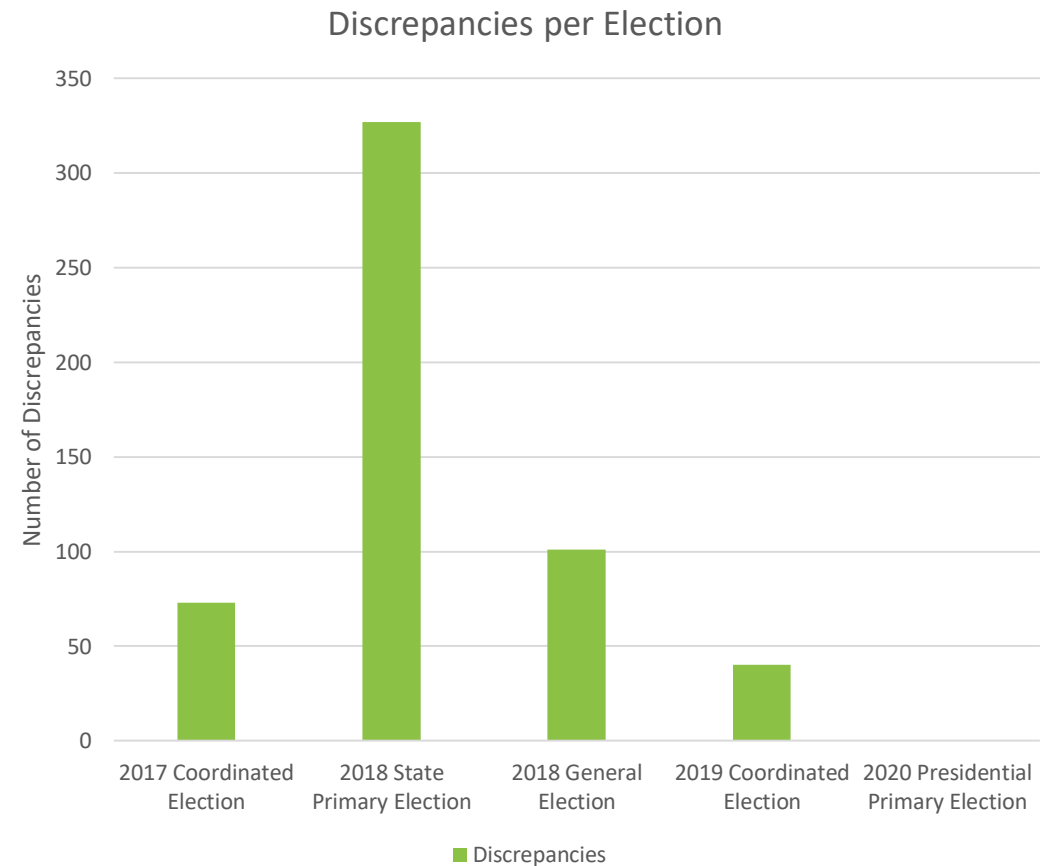
Number of Ballot Cards Audited

The number of ballot cards audited is initially based on the margin of the race, and the total number of cards that contain that race. Discrepancies may require more cards to be audited to confirm the outcome of the reported result. If a high number of ballot cards were audited, it may mean there were many discrepancies reported, or the margins of the races audited were very close.



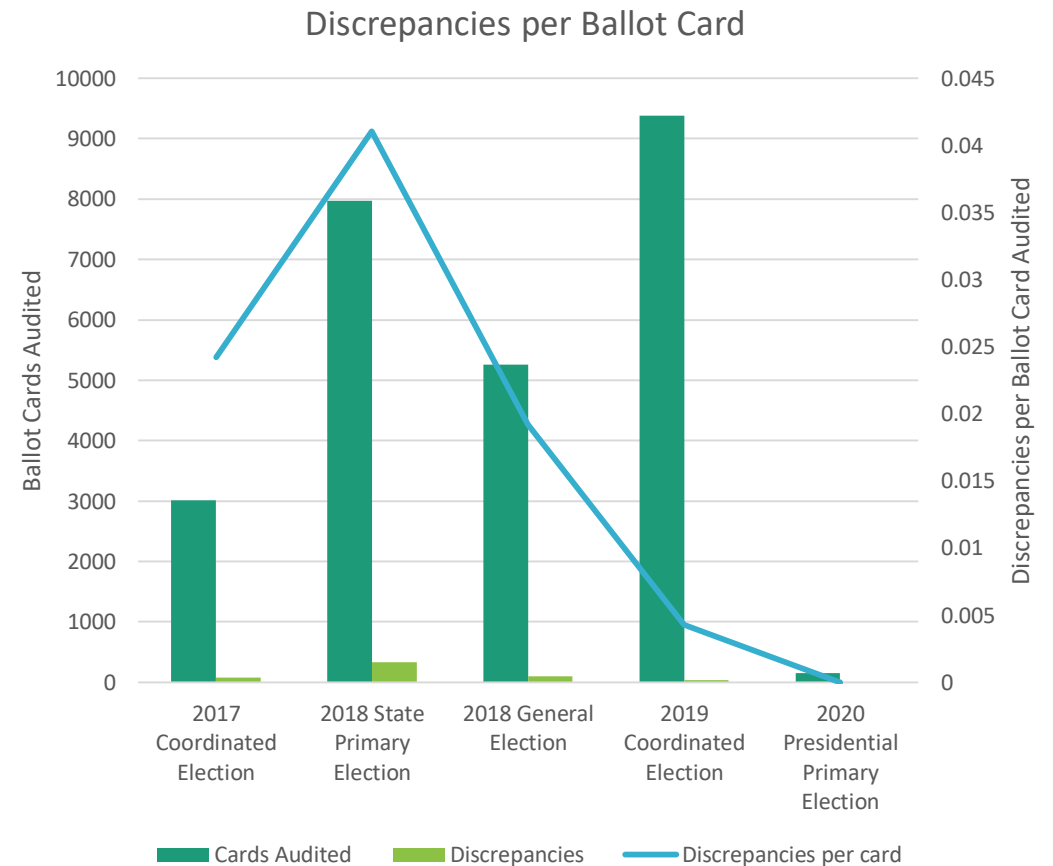
Total Number of Discrepancies

A discrepancy occurs when the audit board examines a ballot that is being audited and reports a result that is different than how the voting system tabulated the ballot. Discrepancies do not necessarily mean the tabulation by the voting system was incorrect, but instead can mean there was a problem with reporting the ballot during the audit.



Discrepancies per Card

The best way to visualize the progress that has been made when performing RLAs is to measure the number of discrepancies reported per ballot card audited. Since the 2018 Primary Election, the number of discrepancies per ballot card has declined. In the 2020 Presidential Primary Election zero discrepancies were reported. (The number of ballots audited was also much lower because of the nature of the election.)



Discrepancies

When the result on the ballot card reported by the audit board and the tabulation by the voting system disagree, it results in a discrepancy. If the discrepancy makes the reported outcome appear more in doubt (the loser received more votes than reported), additional ballots may be required to be audited in the next round to confirm the outcome of the race.

It is important to remember that not all discrepancies are due to a voting system tabulation error. In fact, in the three years that Colorado has been conducting RLAs, our post audit investigation has not identified a single instance of a voting system incorrectly tabulating a ballot.

Types of Discrepancies

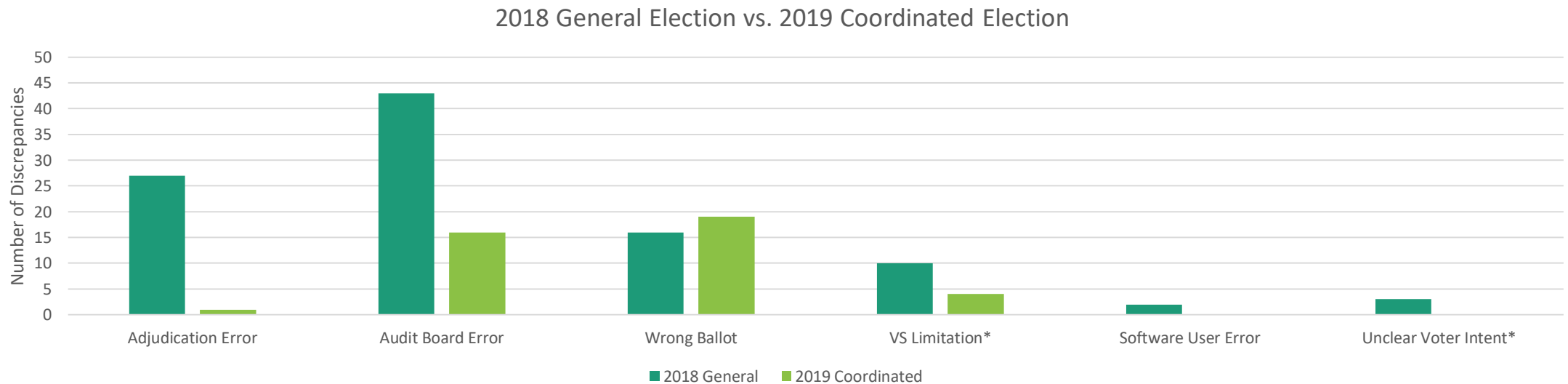
Over the course of three years and five audits, different types of discrepancies have been discovered, which can be sorted into general categories. Explanations for the categories below can be found at the end of the slide show or by clicking on the category.

- [Adjudication Error](#)
- [Audit Board Error](#)
- [Wrong Ballot](#)
- [Voting System Limitation](#)
- [Software User Error](#)
- [Unclear Voter Intent](#)
- [Incorrect Tabulation*](#)

*Has never been reported

Types of Discrepancies, 2018 vs. 2019

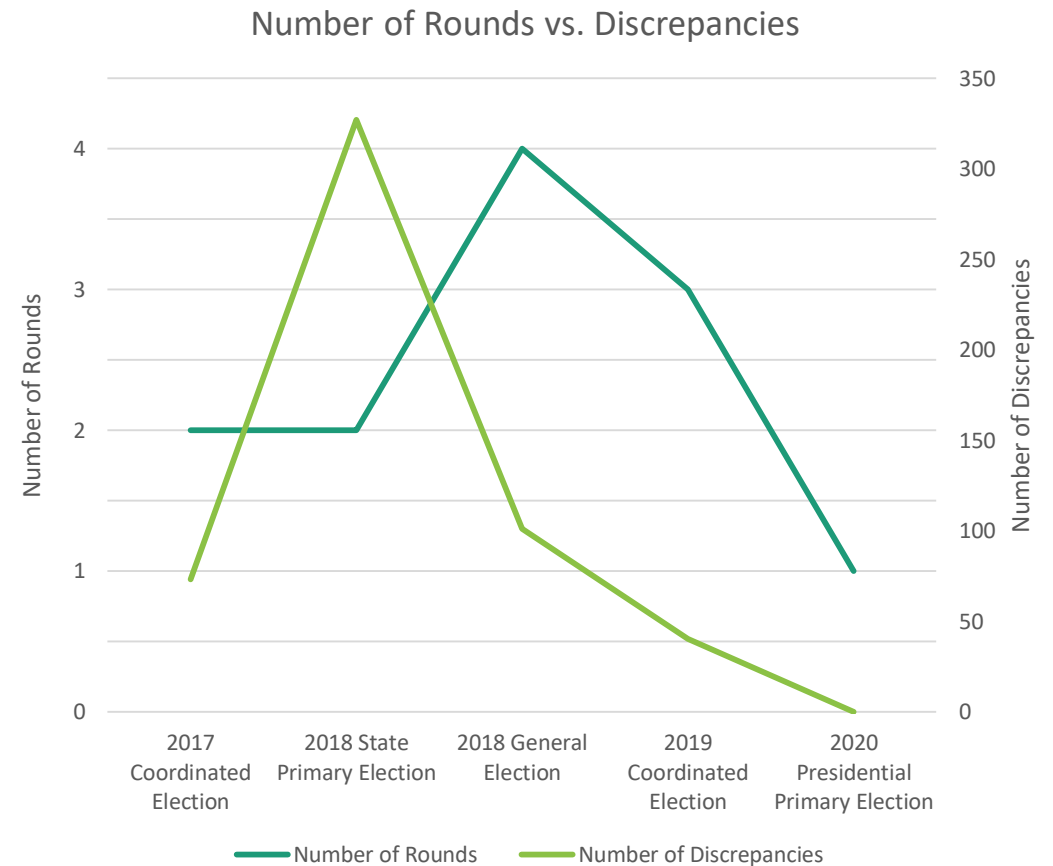
One of the main reasons that the total number of discrepancies dropped between the 2018 General Election and 2019 Coordinated Election is because election judges made fewer mistakes during adjudication, and audit boards were more comfortable using the software to audit ballots. (Data for types of discrepancies prior to the 2018 General Election does not exist.)



Number of Rounds

While the number of rounds to complete the audit does not exactly reflect the decline in the discrepancies, there has been a downward slope since the 2018 General Election.

Note: One round is the fewest number of rounds possible.



Conclusions

As the RLA has become a routine part of elections in Colorado, counties have conducted their audits with more accuracy and efficiency. For jurisdictions thinking of undertaking RLAs or citizens interested in the practice of RLAs, this data should provide a good insight into how they change over time, and what issues should be planned for at the outset.

For a full suite of data for each audit, please visit the [Audit Center](#) at the Colorado Secretary of State's Website.

Types of Discrepancies

Links from a previous slides

Adjudication Error

An error made by the adjudication judges at the time of tabulation, according to the Colorado Secretary of State's Voter Intent Guide. The audit board interpreted the voter intent correctly.

[Back to types of discrepancies](#)

Audit Board Error

The audit board reported the incorrect result into the software. This can occur because of a mis-click or because of a the audit board misinterpreted voter intent.

[Back to types of discrepancies](#)

Wrong Ballot

The audit board did not examine the ballot that the software instructed them to audit. We can infer this if many of the races have clear discrepancies on a ballot, based on ballot images examined by the state after the audit.

[Back to types of discrepancies](#)

Voting System Limitation

In some cases, the voting system cannot detect voter intent, nor does it out stack a ballot for adjudication. These are know instances where the voting system does not tabulate correctly, but it is still working as intended.

For example, if a voter decides to change a vote by crossing out the filled in target area but does not select another choice, the voting system will interpret that mark as a proper vote for a candidate.

In cases where the voter selects a new choice, the system out stacks the ballot for adjudication because it is read as an overvote and the adjudication judges determine voter intent.

[Back to types of discrepancies](#)

Software User Error

The user selected the improper choice in the software. For example, the user selects “ballot card not found” when they can locate the ballot.

[Back to types of discrepancies](#)

Unclear Voter Intent

The Colorado Secretary of State's Voter Intent Guide does not address the voter intent issue encountered on the ballot and the adjudication judges and audit board interpreted it differently.

[Back to types of discrepancies](#)

Incorrect Tabulation

The voting system unexplainably tabulated the ballot the wrong way.

NOTE: This type of discrepancy has never occurred In any of the RLA's over the last three years. This sort of error would raise major red flags about the outcome of the election and would inevitably lead to a deeper investigation into the integrity of the election.

[Back to types of discrepancies](#)